

Sushit Lal Pakrashy

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Summary

AI Engineer building LLM agent systems and the infrastructure that keeps them reliable — published two open-source Python packages implementing peer-reviewed research, each backed by 200+ automated tests and CI/CD. Experience spans multi-agent pipelines (LangGraph, RAG) and active open-source contribution to LiteLLM. Seeking a Summer AI Engineering internship.

Education

Guru Nanak Institute of Technology, Kolkata

Expected 2027

B.Tech, Electronics and Communication Engineering — CGPA: 8.2/10

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript

AI/LLM: LangChain, LangGraph, LiteLLM, RAG, Vector Search (FAISS, ChromaDB), Hugging Face, OpenAI API, prompt-injection detection, agent evaluation, decoding-time monitoring

Infrastructure: FastAPI, REST APIs, Async Python, Docker, GitHub Actions (CI/CD), Redis, Prometheus, Grafana, Observability, Linux/WSL2

Databases: PostgreSQL, Neo4j, MySQL, MongoDB

Testing: pytest, Evaluation Frameworks, Git

Experience

AI/ML Engineer Intern — iNeuBytes

29-06-2026 – 29-07-2026

- Building and evaluating CNN and LSTM models for image classification (CIFAR-10) and sentiment analysis, comparing custom architectures against classical baselines.
- Developing a full-stack ML/NLP web application with a Flask backend and public deployment.

Data Science & AI Intern — Thiranex

23 May 2026 – 22 Jun 2026

- Built an XGBoost heart-disease risk model (~90% accuracy, 0.93 AUC-ROC) with SHAP explainability.
- Built a Random Forest content classifier on Netflix metadata (85.7% accuracy).

Projects

SOPVM — Capability-Gated Runtime for LLM Agents

[GitHub](#) | PyPI: sopvm

Python

- Implemented a research paper as a deterministic YAML-DSL compiler and capability-gated runtime; 15 milestones, 268 tests, adversarial security suite.
- Found and patched an IR-tampering exploit from unvalidated capability metadata.
- Published to PyPI with a public API compatibility checker; verified via clean-env install.

cusum-watch — Calibrated Drift Monitor for Quantized LLMs

[GitHub](#) | PyPI: cusum-watch

Python

- Implemented an e-CUSUM statistics paper as a LiteLLM proxy hook flagging off-distribution generation in quantized models.
- Built two-sided CUSUM detection after single-sided monitoring missed one failure direction.
- Validated false-alarm rate on real sequential data; 79 tests, tiered CI, Prometheus/Grafana metrics, PyPI release.

langgraph-replay — Agent Debugging & Observability Tool

[GitHub](#)

Python, LangGraph, SQLite

- Built a record/replay/blame tool for LangGraph agents with a structural blame engine and LLM-based root-cause diagnosis citing exact source lines.
- Added semantic state diffing, a regression watchdog, and cross-provider latency/cost benchmarking.
- Shipped a Textual TUI debugger and CLI; validated across 141 tests over two major iterations.

SENTINEL — Multi-Module Digital Safety AI Platform

[GitHub](#)

Python, LangGraph, Neo4j, OpenCV Solo-built a 4-module hackathon platform (scam detection, counterfeit-currency analysis via OpenCV, fraud-graph intelligence via Neo4j) with a shared ChromaDB correlation layer linking signals across modules — built for ET AI Hackathon 2.0.

Open Source

- **LiteLLM:** Fixed unconditional cache writes and a TPM-accounting bypass in the parallel request limiter; PR passed 50/50 CI checks (under review).
- **marimo:** Fixed a reactive-error-handling bug in ScratchCellListener (PR #9302).

Achievements

Rank 1, Python & SQL (HackerRank) | Circuitathon 2025 — Team Lead, 1st Runner-up | MCKV Prayas — 2nd Runner-up